INCH-POUND

MIL-C-44112C
18 April 1989
SUPERSEDING
MIL-C-44112B
16 December 1987

MILITARY SPECIFICATION

CRACKERS, FOR MEAL, READY-TO-EAT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two crackers in a flexible bag for use as a component of Meal, Ready-to-Eat, Individual.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

SPECIFICATIONS

FEDERAL

L-C-110 - Cellophane (Coated and Noncoated) Regenerated Cellulose Film

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8920

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L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
QQ-A-1876	-	Aluminum Foil
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-843	-	Cushioning Material, Cellulosic
PPP-C-1797	-	Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam

STANDARDS

FEDERAL

FED-STD-595	-	Colors

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies are available from the Office of Drinking Water Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

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ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AMERICAN OIL CHEMISTS SOCIETY

Official and Tentative Methods of the American Oil Chemists Society

(Application for copies should be addressed to the American Oil Chemists Society, 508 South Sixth Street, Champaign, IL 61820.)

AMERICAN PHARMACEUTICAL ASSOCIATION

The National Formulary

(Application for copies should be addressed to the American Pharmaceutical Association, 2215 Constitution Ave., N.W., Washington, DC 20037.)

ASSOCIATION OF VITAMIN CHEMISTS

Methods of Vitamin Assay

(Application for copies should be addressed to the Interscience Publishers, a division of John Wiley & Sons, 605 Third Ave., New York, NY 10016.)

NATIONAL ACADEMY OF SCIENCE

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Suite 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1238 - Flow Rates of Thermoplastics by Extrusion Plastometer
D 1505 - Density of Plastics by Density Gradient Technique
F 88 - Seal Strength of Flexible Barrier Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Wheat flour. Wheat flour shall be of the commercial type known as cracker flour. It shall be bleached or unbleached and enriched or unenriched. If unenriched flour is used, equivalent enrichment shall be incorporated by the manufacturer at the time of production of the finished product.

3.2.2 Shortening. The shortening shall be hydrogenated cottonseed, soybean, or peanut oil, or any combination thereof. The shortening shall contain anti-oxidants such as BHA, BHT, propyl gallate, or TBHQ singly or in combination to the maximum total amount permitted by FDA regulations. The shortening shall conform to the following requirements when tested in accordance with the indicated methods of the Official and Tentative Methods of the American Oil Chemists Society.

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	Requirement	Method
Color (Lovibond)	2.5R maximum	Cc13b-45
Free fatty acid (FFA)	0.05 percent maximum	Ca5a-40
Fat stability (AOM)	100 hours (after anti-oxidant is added)	Cd12-57
Peroxide (PV)	1.0 mEq oxygen/kg fat maximum (shall not exceed 60 days before use)	Cd8-53
Solid fat index (SFI)		Cd10-57
at 50°F	36 ± 5 percent	
at 70°F	24 ± 2 percent	
at 92°F	10 ± 5 percent	

3.2.3 Yeast, dried, inactive. The dried yeast shall be either primary grown, properly debittered brewers or torula yeast. It shall be mild in flavor, be finely powdered, and comply with the National Formulary quality, except for vitamin potency. Sufficient quantities of fortifying agents as determined by the Methods of Vitamin Assay of the Association of Vitamin Chemists shall be added to the yeast during processing to yield the following potencies:

Micrograms per gram

Thiamine	Not less than 1200
Riboflavin	Not less than 600
Niacin	Not less than 1000
Pyridoxine	Not less than 600

3.2.4 Salt. Salt shall be white, refined sodium chloride with or without anticaking agents. Iodized salt shall not be used.

3.2.5 Leavening yeast. Leavening yeast shall be either compressed bakers or active dry yeast. The yeast shall be a culture of Saccharomyces cerevisiae.

3.2.6 Baking soda. Baking soda shall comply with the requirements of the Food Chemicals Codex. It shall be of a granulation commonly used in the cracker industry and shall be free flowing.

3.2.7 Water. Water used for formulation, ice making, and washing shall conform to the National Primary Drinking Water Regulations.

3.2.8 Malt preparation. Malt preparation shall be derived from water extract of barley malt, dried or evaporated, with or without cereal adjunct, or from malted barley flavor and a fine dextrose hydrate, with or without cereal adjunct.

3.2.9 Protease enzyme. Protease enzyme shall be of high commercial grade and produced from either fungal or bacterial action.

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3.2.10 Yeast food. Yeast food shall be of high commercial grade.

3.2.11 Calcium carbonate. Calcium carbonate shall comply with the requirements of the Food Chemicals Codex.

3.3 Preparation and processing.

3.3.1 Cracker formulation. The crackers shall be prepared by the sponge-dough or straight dough method from ingredients in the following proportions:

Ingredients	Pounds
Wheat flour 1/	200
Shortening	28
Dried yeast, inactive	3
Salt	1-1/2
Leavening yeast	As required
Baking soda	As required
Water	As required
Malt preparation, protease enzyme, yeast food (if necessary)	As required
Calcium carbonate ($CaCO_3$)	2/

1/ Ground crackers derived from clean but broken or otherwise defective crackers from previous runs of this formulation may be added to each batch in an amount not to exceed 5 percent of the flour ingredient.

2/ Add calcium carbonate at a rate to yield 2.5 grams per pound of finished crackers.

3.3.2 Cracker forming and baking. The dough shall be formed, laminated, cut into units, and uniformly well baked so as to yield 4 inches by 4 inches square crackers with a tolerance of ± 1/4 inch on any side. The thickness of the crackers shall be such that a stack of seven crackers (randomly selected and with every other cracker inverted when stacked) will be 1 1/2 inches -1/8 inch +1/4 inch in height. The height measurement shall be taken from the highest point of the stack. The number of baked crackers per pound shall not exceed 24. Each cracker shall have deep docker holes (visible holes extending completely through the baked cracker) placed regularly throughout its surface, approximately 1/4 inch apart. Each of the four sides of each cracker shall be scored once beginning on the outside edge (midpoint) and extending 1/2 inch in length toward the center of the cracker. Each cracker when broken along the partial scorelines shall form four 2 by 2 inch squares.

3.4 Finished product requirements. The crackers shall be uniformly well baked but shall not be excessively brown in color, and shall be flat. The crackers shall be clean, reasonably tender and crisp, but not hard, and shall show no indication of mold or foreign matter. Crackers shall have clean cut

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edges and shall be reasonably free from dusting flour. The crackers shall be free from any foreign or objectionable taste and odor such as musty, rancid, burnt, stale, alkaline, soapy, or excessively salty.

3.4.1 Moisture content. The average moisture content of the finished product shall be not more than 3.5 percent nor less than 2.0 percent. No individual sample shall have a moisture content of more than 4.0 percent or less than 1.5 percent. Testing shall be as specified in 4.5.13.

3.4.2 pH (hydrogen-ion activity). The pH of the finished cracker shall be not less than 6.5 nor more than 8.0 when tested as specified in 4.5.13.

3.4.3 Palatability. The product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.5 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110 "Current Good Manufacturing Practice in Manufacturing, Processing, Packaging or Holding of Human Foods", and the plant sanitation requirements of the appropriate Government inspection agency.

3.6 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by USDA. The USDA will determine the degree of inspection necessary to assure compliance with the requirements of this specification.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

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4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.1.2 Laminated bag material certification. A certificate of compliance may be acceptable as evidence that the characteristics listed below conform to the specified requirements.

Material requirement	Requirement paragraphs	Test procedure
Thickness of films	5.1.1.1 and 5.1.1.2	As specified in L-P-378, except that a machinists' micrometer may be used provided that its graduations and accuracy conform to the requirements of L-P-378.
Aluminum foil thickness	5.1.1.1 and 5.1.1.2	As specified in QQ-A-1876.
Laminated material identification and construction	5.1.1.1 and 5.1.1.2	Laboratory evaluation.
Color of laminated material	5.1.1.1 and 5.1.1.2	Visual evaluation.

4.5.1.3 Unfilled preformed bag seal strength testing. The unfilled bags shall be tested for seal strength in accordance with ASTM F 88, except that the specimen holding clamps shall be spaced 2 inches apart prior to testing and the testing speed shall be 10 or 12 inches per minute. Machines that apply the

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tensile load to the test specimen by movement of the upper or lower clamp may be used. Test specimens shall be cut 1/2 or 1 inch in width and to a length suitable for proper mounting. The lot size shall be expressed in bags. The sample size shall be the number of bags indicated by inspection level S-1. Three adjacent specimens shall be cut from each of the three sealed sides of each bag in the sample. The results shall be reported to the nearest 0.1 pound. The average seal strength of each seal shall be calculated by averaging the strengths of the three test specimens cut from that seal. Any test specimen failing to meet the individual test specimen seal strength requirement or any seal failing to meet the average seal strength requirement specified in 5.1.1.1.1 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.1.4 Preformed bag integrity and air tightness of manufacturer's seals testing. Bag integrity and air tightness shall be determined by placing crackers, fiberboard, or similar rigid material approximately 4 inches by 4 inches by 3/8 inch into the bag and sealing under a vacuum of not less than 15 inches of mercury. Not less than 96 hours after vacuum sealing, the bag shall be examined for vacuum loss in accordance with the procedure in 4.5.4. The lot size shall be expressed in units of unfilled preformed bags. The sample unit shall be one bag. The inspection level shall be I and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 0.65 for major defects.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, bag filling, sealing, and packing requirements. Any nonconformance revealed by actual examination or by review of records of formulation or other valid documents shall be cause for rejection of the involved product.

4.5.2.1 In-process examination at the assembly plant. Bulk-packed crackers shall be examined 72 hours or less before beginning of unit packing into 2-cracker unit packs. Any sample unit with objectionable taste or odor (for example, musty, rancid, burnt, stale, alkaline, soapy, etc.) or any sample unit not reasonably tender and crisp shall be scored as a major defect. The lot size shall be expressed in cases. The sample unit shall be two crackers, randomly selected from each case. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 0.65 for major defects.

4.5.3 Filled and sealed laminated bag examination (unit packs shall be opened). The filled and sealed preformed and form-fill-seal laminated bags shall be examined for the defects listed in table I. The lot size shall be expressed in bags. The sample unit shall be one filled and sealed bag. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 0.65 for major defects and 2.5 for minor defects.

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TABLE I. Filled and sealed bag defects 1/

Category		Defect
Major	Minor	
101		Not size specified
102		Not heat sealed as specified
103		Bags not material specified
104		Not formed as specified
105		Not clean 2/
106		Tear, hole, or open seal
107		Closure seal not produced by specified method
108		Evidence of delamination 3/
109		Seal width not as specified
110		Bag has foreign odor
111		Required labeling missing, incorrect, illegible, or that smudges
112		Color not as specified
113		One or more crushed crackers 4/
114		Presence of entrapped matter (e.g., cracker crumbs, flour, etc.) that reduces the effective closure seal to less than 1/16 inch 5/
115		Presence of flex fractures in the aluminum foil 6/
116		More or less than two crackers per bag
	201	Closure or top seal extends into or below tear notch 7/
	202	Tear notch or serrations missing
	203	Tear notch or serrations not located as specified

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TABLE I. Filled and sealed bag defects 1/ (cont'd)

Category		Defect
Major	Minor	
	204	Depth of tear notch or serrations not as specified
	205	One or more broken crackers 8/

1/ Any evidence of insect or rodent infestation shall be cause for rejection of the lot.

2/ Outer packaging shall be free from foreign matter which is unwholesome, has the potential to cause bag damage (for example, glass, metal filings, etc.) or generally detracts from the clean appearance of the package. The following examples shall not be scored as defects for unclean:

a. Foreign matter which presents no health hazard or potential bag damage and which can be readily removed by gently shaking the package or by gently brushing the package with a clean dry cloth.

b. Dried product which affects less than 1/8 of the total surface area of one bag face (localized and aggregate).

c. Water spots.

3/ Delamination shall be scored as a defect except delamination of outer ply when located in the seal area 1/16 inch or further from food product edge of seal. Bags exhibiting this type of delamination shall be tested by manually flexing the delaminated area 10 times. The area of delamination shall be held between the thumb and forefinger of each hand with both thumbs and forefingers touching each other. The delamination area shall then be rapidly flexed by rotating both hands in alternating clockwise - counterclockwise directions. Care shall be exercised when flexing delaminated area near the tear notches to avoid tearing the bag material. After flexing, the separated outer ply shall be grasped between the thumb and forefinger and gently lifted toward the food product edge of the seal. If the separated area is too small to be held between thumb and forefinger, a number two stylus shall be inserted into the delaminated area and gentle lifting force applied against the outer ply. If separation of the outer ply can be made to extend to less than 1/16 inch from the product edge of the seal with no discernible resistance to the gentle lifting, it shall be considered evidence of delamination.

4/ Crushed crackers is any unit where more than 1/8 of cracker volume is destroyed.

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5/ The effective closure seal is defined as any uncontaminated, fusion bonded, continuous path, minimum 1/16 inch wide from side seal to side seal that produces a hermetically sealed bag.

6/ To examine for flex fractures, the inside surface of each bag side shall be placed over a light source and the outside surface observed for the passage of light. Observation of light through the bag material in the form of a curved or straight line shall be evidence of the presence of flex fractures in the aluminum foil.

7/ Applicable to preformed bags only.

8/ Broken into three or more pieces, other than along cracker serrations.

4.5.4 Laminated bag vacuum examination. The filled and sealed bag shall be visually examined for a proper vacuum level not less than 96 hours after filling and sealing. The sealed bag shall continue to exhibit tight adherence to the large flat surfaces of the contents when a pulling force is applied at the center of each side seal. This force shall be applied by holding each side seal between thumb and forefinger of each hand, simultaneously exerting a slight pull with both hands. Any evidence of loss of vacuum shall be considered a major defect. The lot size shall be expressed in bags. The sample unit shall be one filled and sealed bag. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 0.65.

4.5.5 Product examination to be performed when unit packed in laminated bags as specified in 5.1.1 at the cracker manufacturer's plant only. The finished product shall be examined for the defects listed in table II. The lot size shall be expressed in bags. The sample unit shall be one filled and sealed bag. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 10.0 for minor defects.

TABLE II. Product defects (laminated bag unit pack) 1/ 2/

Category		Defect
Major	Minor	
101		Not uniformly well baked
102		Excessively brown in color (for example, dark brown, toasted, scorched)
103		Not reasonably tender and crisp 3/
104		Hard (for example, brittle, difficult to bite into) 3/
105		Objectionable taste or odor (for example, musty, rancid, burnt, stale, alkaline, soapy, excessively salty) 3/

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TABLE II. Product defects (laminated bag unit pack) 1/ 2/ (cont'd)

Category		Defect
Major	Minor	
106		More than 24 crackers per pound 4/
107		Height of seven crackers more than 1 3/4 inches 5/
	201	Edges not clean cut
	202	Not reasonably free from dusting flour (white powdery appearance on surface due to dusting powder)
	203	Length or width of cracker is less than 3-3/4 inches, or more than 4-1/4 inches 3/
	204	Height of seven crackers less than 1-3/8 inches 5/
	205	Bag contains broken or crushed crackers

1/ The presence of foreign material (for example, glass, dirt, insect, insect parts, hair, wood, metal) or mold shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability or overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by a Federal Grain Inspection Service Agent (FGIS).)

3/ Examine one cracker per sample unit.

4/ Select 24 crackers for this examination.

5/ Select crackers from those used for defect 106.

4.5.6 Laminated bag closure seal testing. The filled and sealed bags shall be tested in accordance with ASTM F 88, except that the specimen holding clamps shall be spaced 2 inches apart prior to testing and the testing speed shall be 10 or 12 inches per minute. Machines that apply tensile load to the specimen by movement of the upper or lower clamp may be used. The test specimens shall be cut to a length suitable for proper mounting. For preformed bags, three adjacent specimens, 1/2 or 1 inch wide shall be cut from the closure seal of each bag in the sample. For the form-fill-seal bags, three adjacent specimens, 1/2 or 1 inch wide shall be cut from each side and each end of each bag in the sample. For the preformed bag, the average seal strength of the closure seal shall be calculated by averaging the test results of the three test specimens cut from that seal. For the form-fill-seal bag, the average seal strength of each side and end of the bag shall be calculated by averaging the test results

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of the three specimens cut from that side or end. The results shall be reported to the nearest 0.1 pound per inch of width. Any test specimen or average seal strength failing to meet the requirements of 5.1.1.1.2 or 5.1.1.2.1 shall be scored as a major defect. The lot size shall be expressed in bags. The sample unit shall be one filled and sealed bag. The sample size shall be the number of bags indicated by inspection level S-1 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

4.5.7 Product examination (when bulk packing as specified in 5.1.2.1 is utilized). The product shall be examined for the defects listed in table III. The lot size shall be expressed in cartons or boxes. The sample unit shall be three crackers. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 10.0 for minor defects.

TABLE III. Product defects (bulk pack) 1/ 2/

Category		Defect
Major	Minor	
101		Not uniformly well baked
102		Excessively brown in color (e.g., dark brown, toasted, scorched)
103		Not reasonably tender and crisp 3/
104		Hard (e.g., brittle, difficult to bite into) 3/
105		Objectionable taste or odor (e.g. musty, rancid, burnt, stale, alkaline, soapy, excessively salty) 3/
106		More than 24 crackers per pound
	201	Edges not clean cut
107		Height of seven crackers more than 1 3/4 inches 4/
	202	Not reasonably free from dusting flour (white powdery appearance on surface due to dusting powder)
	203	Height of seven crackers less than 1 3/8 inches 4/
	204	Length or width of cracker is less than 3 3/4 inches or more than 4 1/4 inches 4/

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1/ The presence of foreign material (for example, glass, dirt, insect, insect parts, hair, wood, metal) or mold shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability or overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an FGIS agent.)

3/ Examine one cracker per sample unit.

4/ One determination per sample unit shall be made for size and height.

4.5.8 Product condition examination. Crackers bulk packed for shipment to ration assemblers shall be examined for the defects listed in table IV. The lot size shall be expressed in cartons or boxes. The sample unit shall be one carton or box. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 0.65 for major defects.

TABLE IV. Cracker condition defects (3-pound carton or 18 pound box)

Category	Defect
Major	
101	More than eight 4 by 4 inch crackers separated along a score line, broken, or crushed 1/
102	More than fourteen 4 by 4 inch crackers containing checking 2/

1/ A separated, broken, or crushed cracker is one that is in two or more distinct pieces or has more than 1/16 of its area missing.

2/ Checking is the appearance of one or more fracture lines on the surface.

4.5.9 Carton (3-pound) external examination. The cartons bulk packed with 3 pounds of crackers shall be examined for the external defects listed in table V. The lot size shall be expressed in units of cartons. The sample unit shall be one filled and wrapped carton. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 0.65 for major defects and 2.5 for minor defects.

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TABLE V. External defects (3-pound cartons)

Category		Defect
Major	Minor	
101		Overwrap(s) missing
102		Overwrap(s) not as specified
103		Labeling, if on carton, not legible through overwrap
104		Nomenclature missing, incorrect, or illegible
	201	Tear, hole, or open seam in both overwraps exposing fiberboard 1/
	202	Other required label information missing, incorrect, or illegible
	203	Net weight more than 2 percent under labeled net weight 2/
	204	Overlap of each sheet less than 1/2 inch at seams and end closures
	205	Tear or hole in either overwrap or open seal in the overwrap 1/
	206	Longer seam of each overwrap not on opposite panels of carton
	207	Not clean

1/ Vent holes shall not be scored as a defect.

2/ Lot shall be rejected if sample data indicated lot average net weight less than labeled net weight.

4.5.10 Carton (3-pound) internal examination. The cartons bulk packed with 3 pounds of crackers shall be examined for the internal defects listed in table VI. The lot size shall be expressed in units of cartons. The sample unit shall be one filled carton. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 4.0 for minor defects.

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TABLE VI. Carton (3-pound) internal defects

Category		Defect
Major	Minor	
101		Liner missing or not fabricated as specified
102		Liner not completely protecting product from interior surfaces of carton
103		Open end of bag liner not closed in the manner specified
104		Liner not clean
105		Liner has foreign odor
	201	Liner stretched due to inadequate size of liner.

4.5.11 Shipping container (18-pound) external examination. Shipping containers bulk packed with 18 pounds of crackers for direct shipment to ration assemblers shall be examined for the external defects listed in table VII. The lot size shall be expressed in units of shipping containers. The sample unit shall be one filled, marked, sealed, and wrapped shipping container. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for minor defects.

TABLE VII. Shipping container (18-pound) external defects

Category		Defect
Major	Minor	
101		Overwrap(s) missing
102		Overwrap(s) not as specified
103		Not constructed as specified
104		Not closed as specified
105		Labeling not legible through overwrap
106		Nomenclature missing, incorrect, or illegible
	201	Other required label information missing, incorrect, or illegible

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TABLE VII. Shipping container (18-pound) external defects (cont'd)

Category		Defect
Major	Minor	
	202	Tear, hole, or open seam in both overwraps exposing fiberboard 1/
	203	Net weight more than 2 percent under labeled net weight 2/

1/ Vent holes shall not be scored as a defect.

2/ The lot shall be rejected if the sample average net weight is less than the sample average marked net weight.

4.5.12 Shipping container (18-pound) internal examination. Shipping containers bulk packed with 18 pounds of crackers for direct shipment to ration assemblers shall be examined for the internal defects listed in table VIII. The lot size shall be expressed in units of shipping containers. The sample unit shall be one filled shipping container. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 4.0 for minor defects.

TABLE VIII. Shipping container (18-pound) internal defects

Category		Defect
Major	Minor	
101		Polyethylene bag missing or not as specified
102		Corrugated parchment bag liner and/or dividers missing or not positioned as specified
103		Tear or hole in polyethylene bag
104		Polyethylene bag not closed with twist tie or similar closing device
105		Polyethylene bag and/or corrugated parchment not clean
106		Corrugated box liner or top and bottom pads missing
107		Polyethylene bag and/or corrugated parchment bag liner has foreign odor

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TABLE VIII. Shipping container (18-pound) internal defects (cont'd)

Category		Defect
Major	Minor	
	201	Polyethylene bag not thickness specified
	202	Crackers not packaged in uniform rows and layers

4.5.13 Moisture and pH testing. The finished product shall be prepared and tested for moisture content in accordance with the Total Solids (Moisture, Indirect Method) and for pH concentration in accordance with the hydrogen-ion Concentration (Electrometric Method) in the Wheat Flour section, Cereal Foods chapter of the Official Methods of Analysis of the Association of Analytical Chemists. The sample unit shall be one filled and sealed laminated bag or two crackers taken from different cartons or boxes, as applicable. The lot size shall be expressed in pounds. The sample size shall be the number of sample units indicated by inspection level S-2. The results shall be reported to the nearest 0.1 percent. A sample with a moisture content more than 4.0 percent or less than 1.5 percent, or an average result of more than 3.5 percent or less than 2.0 percent shall be classified as a major defect and shall be cause for rejection of the lot. Any sample unit having a pH result not within the limits of 6.5 to 8.0 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.14 Shipping container examination (for laminated barrier bags). Examination of the filled and closed shipping containers packed with 2-cracker unit packs shall be in accordance with the appendix of PPP-B-636. In addition, the following defects shall be included in the table of examination: Major: Marking missing, incorrect, or illegible; Minor: Count not as specified (see 5.2.1.3); cushioning material not placed as specified (see 5.2.1.3). Shipping containers containing boxes or cartons shall not be acceptable if markings indicate a total net weight of more than 45 pounds.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or B as specified (see 6.1).

5.1.1 Level A (for ration assembly). Two crackers shall be unit packed in flexible laminated barrier bags as specified in 5.1.1.1 or 5.1.1.2. When crackers are unit packed by the ration assembler, the crackers shall not be more than 90 days old from date of manufacture.

5.1.1.1 Preformed bags. The preformed bags shall be fabricated from 0.002 inch thick (± 0.0002 inch) ionomer film bonded to 0.00035 inch thick (± 0.000035 inch) primed aluminum foil by melt extrusion with 0.001 (± 0.0002 inch) ionomer and subsequently bonding 0.0005 inch thick (± 0.00005 inch) polyethylene

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terephthalate film to the opposite side of the primed aluminum foil by melt extrusion with 0.0007 (+ 0.00014 inch) low density polyethylene. Alternatively, 0.003 inch thick (+ 0.0003 inch) linear low density polyethylene may be used in lieu of the ionomer. The ionomer film shall be the sodium neutralized copolymer of ethylene and methacrylic acid, having a melt index of 1.5 g/10 minutes in accordance with ASTM D 1238 and a density of 0.94 g/cc in accordance with ASTM D 1505. The ionomer used for bonding to the foil shall be the zinc neutralized copolymer of ethylene and methacrylic acid, having a melt index of 5.0 g/10 minutes and a density of 0.94 g/cc. When linear low density polyethylene is used, it shall be the copolymer of ethylene and octene -1, having a melt index range of 0.8 to 1.2 g/10 minutes and a density range of 0.918 to 0.922 g/cc. The exterior (polyethylene terephthalate) surface of the bag shall be colored overall with a color in the range from 34079 through 34087, or 24052 through 24087, or, 10045 or 30045 through 30118 (except that 30109 may not be used) of FED-STD-595. The material shall be suitably formulated for food packaging and shall not impart an odor or flavor to the product being unit packed. The material shall show no evidence of delamination, degradation, or foreign odor when heat sealed or fabricated into bags.

5.1.1.1.1 Bag construction. The bag shall be a flat style bag having inside dimensions of 5-1/2 by 6 inches (+ 1/8 inch), with the 5-1/2 inch dimension being the opening of the bag. The empty bag shall be made by heat sealing all three edges with 3/8 inch (+ 1/8 inch) wide seals. The heat seals shall be made in a manner that will assure hermetic seals and preclude leakage of air into the unit pack through the seals, seal junctions or through defects created when the bags are filled and sealed as specified in 5.1.1.1.2. Integrity of the bag and air tightness of the seals shall be tested in accordance with 4.5.1.4. The side and bottom seals shall have an average seal strength of not less than 6 pounds per inch and no individual specimen shall have a seal strength of less than 5 pounds per inch (see 4.5.1.3). A V-shaped or C-shaped (half round) tear notch at least 1/32 inch deep, located 1 to 1-1/4 inches from the top edge of the bag, shall be made in one or both side seals. The distance between the inside edge of the tear notch and the inside edge of the seal shall be at least 3/16 inch. One side of the open end of the bag may be provided with an extended foldover lip extended not more than 5/16 inch to facilitate opening and filling; if so, tear notch location shall be measured from the top of the bag (excluding the extended or foldover lip).

5.1.1.1.2 Bag filling and sealing. The crackers shall be filled into the bags in a manner to avoid puncturing the material. The filled bag shall be closed under a vacuum of not less than 15 or more than 20 inches of mercury (see 4.5.4) with a 3/8 + 1/8 inch heat seal. If thermal impulse closure seals are used, a seal width of 1/8 inch will be acceptable. The closure seal shall be free of entrapped matter (cracker crumbs, flour, etc.) that reduces the effective closure seal to less than 1/16 inch wide. The closure seal shall not extend below the tear notch on either side of the bag. The closure seal shall have an average seal strength of not less than 6 pounds per inch and no individual specimen shall have a seal strength of less than 5 pounds per inch (see 4.5.6) and shall meet the requirements of the bag integrity test (see 4.5.1.4).

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5.1.1.2 Horizontal form-fill-seal bags. The horizontal form-fill-seal bag shall consist of a formed tray-shaped body with a flat sheet, heat sealable cover or a tray-shaped body with a tray-shaped heat sealable cover. The tray-shaped body, the flat sheet cover and the tray-shaped cover shall be fabricated from a 3-ply flexible laminate barrier material consisting of 0.0009 inch thick (± 0.00009 inch) oriented polypropylene bonded to 0.0007 inch thick (± 0.00007 inch) aluminum foil with 10 pounds per ream pigmented polyethylene or adhesive and bonding the opposite side of the aluminum foil to 0.003 inch thick (± 0.00006 inch) ionomer or a blend of not less than 50 percent linear low density polyethylene and polyethylene. The linear low density polyethylene portion of the blend shall be the copolymer of ethylene and octene-1 having a melt index range of 0.8 to 1.2 g/10 minutes in accordance with ASTM D 1238 and a density range or 0.918 to 0.922 g/cc in accordance with ASTM D 1505. Alternatively, the tray-shaped body and flat-sheet cover combination package may be fabricated from material consisting of 0.0005 inch thick (± 0.00005 inch) polyester laminated to 0.0007 inch thick (± 0.00007 inch) aluminum foil that has 0.003 inch thick (± 0.0006 inch) ionomer, or polyethylene film laminated or extrusion coated to the opposite side. The polyester shall be on the outside of the package. The color requirements of the exterior (orientated polypropylene or polyester side) of the laminate shall be as specified in 5.1.1.1. The laminate shall be suitably formulated for food packaging and shall not impart any odor or flavor to the product being unit packed.

5.1.1.2.1 Bag construction. The tray-shaped body and the tray-shaped cover shall be formed by drawing the flexible laminate material into an appropriately shaped cavity. The flat cover shall be in the form of a flat sheet of the barrier material taken from roll-stock. Two crackers shall be placed into the tray-shaped body of the bag. The filled bag body shall then be hermetically sealed under a vacuum of not less than 15 nor more than 20 inches of mercury (see 4.5.4). Bag closure shall be effected by heat sealing the flat cover or the tray-shaped cover to the tray-shaped body along the entire bag perimeter. The outside dimensions of the sealed bags shall be 5-1/2 inches wide by 5-1/2 inches long (± 1/16 inch). The closure seal width shall be 3/8 ± 1/8 inch. The closure seal shall have an average seal strength of not less than 6 pounds per inch of width and no individual specimen shall have a seal strength of less than 5 pounds per inch (see 4.5.6). A V-shaped or C-shaped tear notch shall be placed on one outside edge or two opposite outside edges of the bag to effect easy opening of the bag in the machine direction of the bag body and cover laminates. Each tear notch shall be located 1/2 inch (± 1/16 inch) from an outside corner of the bag. Each tear notch shall be a minimum 1/32 inch deep, but shall not extend into the seal to a depth that would reduce the seal width to less than 3/16 inch. Alternatively, if the bag has serrated edges, the serrations may be used as tear notches provided that the serrations are located to effect easy opening in the machine direction of the bag laminates, the serrations are sharp (no plastic tailings exist), and the serration depth and the minimum seal width at the serrations are in accordance with the requirements specified for the V or C-shaped notches. The sealed bag shall not show any

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evidence of material degradation, aluminum stress cracking, delamination, or foreign odor when heat sealed or fabricated into bags. The seals shall be free of entrapped matter (cracker crumbs, flour, etc.) that reduces the effective closure seal to less than 1/16 inch wide.

5.1.2 Level B (for shipment to ration assembler).

5.1.2.1 Bulk packing. Crackers shall be bulk packed in a 3-pound carton or an 18-pound box as specified in 5.1.2.1.1 or 5.1.2.1.2 (see 6.1).

5.1.2.1.1 Three-pound carton. Not more than 3 pounds of crackers shall be unit packed, in uniform rows and layers, in a good commercial carton. The carton shall be lined with a waxed, unbleached, glassine paper having a nominal basis weight of not less than 25 pounds per ream (24 by 36-500). The sheet shall be waxed with 3 pounds per ream (24 by 36-500) of fully refined paraffin wax having a melting point of not less than 130°F. The wax shall be odorless and tasteless. The liner shall overlap itself sufficiently to insure adequate protection of the contents from the board. Alternatively, the carton shall be lined with a bag constructed of polyethylene film having a nominal thickness of 0.0015 inch when tested in accordance with the requirements of L-P-378. The bag liner shall be of sufficient size to inclose the crackers without stretching. The bag shall be of sufficient length so that the open end may be closed by pressing opposite sides together and folding the open end with three or more folds to form a moisture proof seal; alternatively, the polyethylene inner liner may be closed with a twist tie closure forming a moisture proof seal. The film shall conform to the requirements of the Federal Food, Drug, and Cosmetic Act, for contact with food and shall not impart any odor or flavor to the product being packed. The filled and closed carton shall then be overwrapped with two sheets of waxed paper, two sheets of cellophane, or a combination of one sheet of waxed paper and one sheet of cellophane. The waxed paper overwraps shall be made from a good quality sulphite of kraft paper having a nominal basis weight of 23 pounds per ream (24 by 36-500), and shall be wet waxed with 14 pounds per ream (24 by 36-500) of a fully refined paraffin wax having a melting point of not less than 130°F. The cellophane overwraps shall be type IV, class 2, having a nominal thickness of 0.0009 inch in accordance with L-C-110. Both sheets shall be large enough to effect an overlap at the seam and end closures of not less than 1/2 inch. Each wrap shall have a continuous heat seal made along the full length of the overlaps. The longer seams of the two sheets shall be located on opposite panels of the carton.

5.1.2.1.2 Eighteen-pound polyethylene bag. Approximately 18 pounds of crackers shall be unit packed in a 0.002 inch thick polyethylene bag. The crackers shall be positioned on edge and arranged in two layers of three rows each. All six rows shall be of equal length and there shall be no slack fill. Each layer and each row shall be separated by a single-faced corrugated parchment pad. The crackers, arranged as specified, shall be approximately 15-3/4 inches long, 12-5/8 inches wide, and 8-5/16 inches high. The crackers shall be separated from the inside surface of the bag by a U-shaped single-faced corrugated parchment sheet, which completely covers the bottom surface and the

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two longest outside surface areas of the arranged crackers. The open end of the bag shall be located at the top of the unit pack. The bag shall be securely closed with a twist-tie or comparable closing device. The polyethylene bag and the single-faced corrugated parchment sheet and pad material shall be suitable formulated for contact with food and shall not impart any odor or flavor to the crackers.

5.2 Packing. Packing shall be level C.

5.2.1 Level C.

5.2.1.1 Eighteen-pound shipping containers. Each eighteen-pound polyethylene bag of crackers, preserved as specified in 5.1.2.1.2, shall be packed in a 275-pound test, C- flute, RSC-L container, approximately 16 by 12-7/8 by 8-5/16 inches, with top and bottom corrugated pads. The box liner and the top and bottom pads shall be made from minimum 200-pound test corrugated fiberboard. The container shall be glue sealed and shall be double shrink wrapped with not less than 0.0015 inch thick polyethylene. Each of the shrink wraps shall be closed on opposite panels of the box. Each wrap shall be provided with vent holes on opposite panels of the box, to allow for the escape of entrapped air.

5.2.1.2 Bulk pack cartons. Not more than 45 pounds of the product, preserved as specified in 5.1.2.1.1, shall be packed in a shipping container in accordance with the Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.2.1.3 Laminated barrier bags. Ninety-six unit packs in accordance with 5.1.1 shall be packed flat in a snug-fitting fiberboard box constructed and closed in accordance with style RSC-L, type CF, class domestic, variety SW, grade 175 of PPP-B-636. Cushioning material not less than 1/8 inch thick in accordance with type II, class B of PPP-C-843 or cushioning material not less than 1/16 inch thick in accordance with PPP-C-1797 shall be placed between each of the layers and at the bottom and the top of the box to minimize movement of individual unit packs.

5.3 Marking.

5.3.1 Unit packs. Unit packs shall be permanently imprinted with the word "CRACKERS" with 3/8 inch high letters in accordance with MIL-STD-129.

5.3.2 Bulk pack boxes and shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129. The 18-pound boxes (see 5.1.2.1.2) shall have the date of cracker manufacture printed on one side and one end of the boxes.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.4, and 6.2).
d. Provisions for approved preproduction samples (see 3.4.3 and 6.2).
e. Selection of applicable level of preservation (see 5.1).
f. When a 3-pound carton or 18-pound box bulk pack is required (see 5.1.2.1).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Cracker size, shape, height and weight. To aid in maintaining correct size and shape of crackers, as well as height, and weight (see 3.3.2), and to hold "blistering" and "checking" to none or to a minimum, the following is listed for guidance:

a. Increase docker pins to 40 or more per cup.
b. Enclose the cooling system.
c. Control cooling conditions from the oven to the packer, to inhibit "checking" and "shattering".

6.4 Subject term (key word) listing.

Biscuit
Operational Ration

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD, TS
Navy - MC
DLA - SS

Preparing activity:

Army - GL

(Project 8920-0522)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-44112C	CRACKERS, FOR MEAL, READY-TO-EAT

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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